Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	21-10-2026
Team ID	23KD1A0571
Project Name	Employee Raise Issue – Record Producer & Service Portal Integration Guide
Maximum Marks	4 Marks

Technical Architecture:

The solution follows the path an issue takes from the employee to the database. The employee lands on the Service Portal, uses the *Raise an Issue* widget to open the Record Producer, the form rules are applied while the form is filled in, and the record is saved in the custom table. Everything sits inside one scoped application on a ServiceNow developer instance.

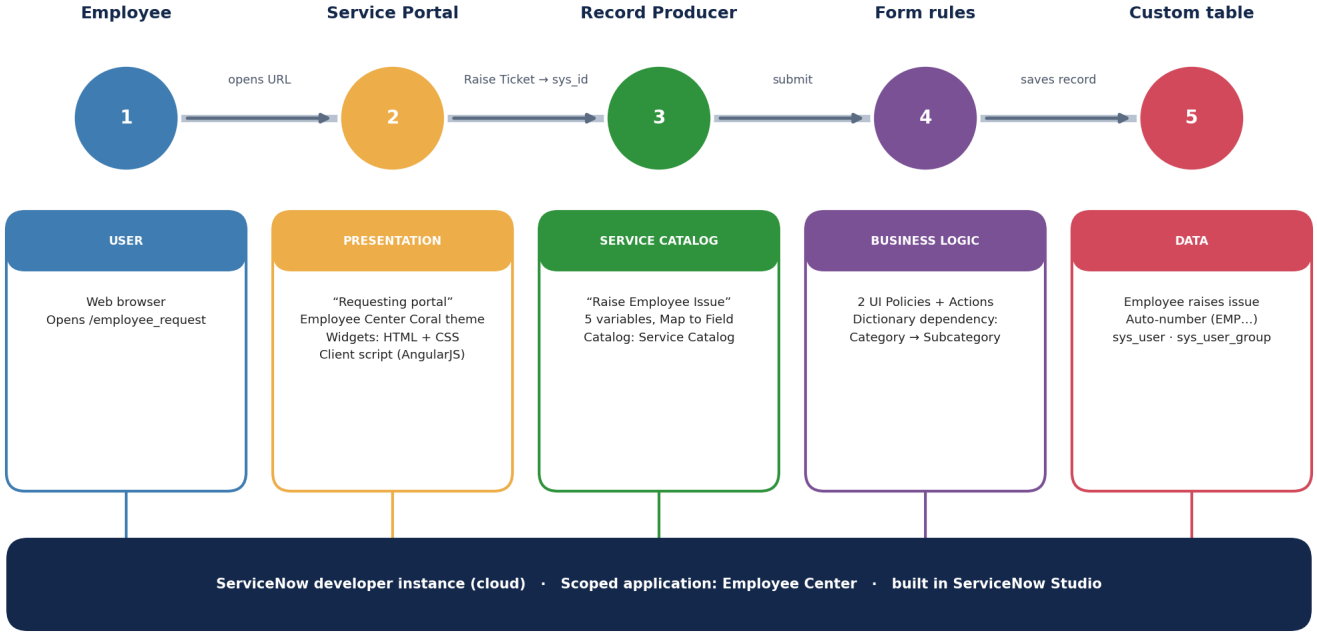


Figure 1 – Technical architecture of the Employee Raise Issue solution

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Service Portal where employees land and start raising an issue	Service Portal “Requesting portal” (employee_request), Employee Center Coral theme, HTML, CSS
2	Application Logic-1	“Raise an Issue” widget that takes the employee to the form	AngularJS client controller ($scope, $location), redirect using the Record Producer sys_id
3	Application Logic-2	Guided form that turns the employee’s answers into a record	Service Catalog Record Producer with five variables mapped to table fields
4	Application Logic-3	Rules on the form: short description mandatory, number read-only, Subcategory options based on Category	UI Policies, UI Policy Actions, Dictionary dependent field and Choices
5	Application Container	Keeps all the project artefacts together in one place	Scoped application “Employee Center”, built in ServiceNow Studio (App Engine)
6	Database	Stores every issue with its auto-generated number	ServiceNow table “Employee raises issue” (Number, Requester, Category, Subcategory, Short description, State, Priority, Assignment group)
7	Reference Data	Requester and Assignment group are picked from existing system tables	sys_user, sys_user_group
8	Cloud Platform	Hosts and runs the application	ServiceNow developer instance (cloud-hosted)
9	File / Media Storage	Images used on the catalog form and the portal	Record Producer picture, portal logo (uploaded images)
10	External Resource	Background image of the Raise an Issue widget	Image loaded from an Unsplash URL through CSS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Client-side Scripting & Styling	Animated widgets (floating heading, staggered list, soft glow) and a click handler that opens the form	CSS3 animations, AngularJS client script
2	Data Consistency	Fixed choice values and dependent subcategories keep records uniform	Choice fields, Dictionary dependency
3	Input Validation	Short description is required when State is New; the Number field cannot be edited	UI Policies, UI Policy Actions
4	Maintainability	All artefacts live in one scoped application; the two widgets are separate portal components	Scoped application, Service Portal widgets
5	Access	The portal uses the platform login page and the Employee Center menu	Service Portal configuration (login page, main menu)
6	Usability	Reuses the Employee Center homepage layout and theme; one click from the widget to the form	Employee Center Coral theme, ec_standard_home

References:

- https://developer.servicenow.com
- https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html
- https://www.servicenow.com/community/